

A Single-Phase to Ground Fault Location Method Based on Least-Squares Parameter Estimation in Arc-Suppression-Coil Grounding System

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Abstract—The accuracy of the distribution network fault location is important to guarantee the safety and reliability of the power system. There are many achievements in the research of fault location technology for the distribution network, but it is still difficult to locate fault of neutral point via arc suppression coil grounding system accurately in the application of single-phase to ground fault line selection or detection, because of the problems of a lack of fault selection components, inaccurate line parameters and expensive phase angle information collection devices. This paper proposes a fault location method of single-phase to ground fault based on least-squares parameter estimation. Taking the zero-sequence equivalent circuit model of single-phase ground fault as the basis, regarding the fault distance, fault resistance and inaccurate line parameters as the unknown parameters, and treating the time domain relationship between the fault zero-sequence voltage and current reflected through the zero-sequence equivalent circuit as fault location equation, the optimum solution of the distance from the fault point to the busbar can be calculated with the least-square parameter estimation method. In addition, the method takes the unknown nature of the parameters into account, and only need to measure the fault three-phase voltage and three-phase current data for determining the fault location without phase angle information. The validity of the method was verified by simulation experiments.

Keywords—single-phase to ground fault, fault location, parameter estimation, least-squares algorithms

I. INTRODUCTION

In recent years, China's power grid is gradually developing towards intelligence and digitalization, with the proposal of strong smart grid and the development of digital twin technology. Smart distribution grids are becoming increasingly structured and functional. Fast and accurate fault location in distribution network is the basic requirement and the important function of smart grid, which plays an important role in ensuring the safe, stable and economic operation of power system.

Among the various types of faults in the grid, the highest probability of occurrence in the distribution network is single-phase to ground faults, accounting for more than 80% of the overall fault types. These faults are mostly grounded through non-metallic media such as tree branches. The transitional resistance exhibits a distinctly high resistance state with a weak fault current. As a result, it is very difficult to localize the point of failure. Domestic and foreign scholars have done a lot of research on this issue, but accurate fault location methods are still in the research stage. The existing methods mainly include traveling wave analysis^[1-3], impedance method^[4-8] and parameter estimation method^[9-12].

Although the traveling wave method is less affected by the transition resistance, the type of fault and the mode of operation of the system^[2], the wave measurement relies on a specialized detection device. At the same time, the short length of distribution network lines and the many branch nodes make traveling wave identification difficult. Installation of the unit increases operating costs which are not economical. The impedance method utilizes the current-voltage characteristics during a fault to create a fault location equation to solve. With a simple principle and good economy, it is more suitable for fault location in distribution networks. Depending on the electrical quantity collected by the measurement, it can be categorized into double-ended fault location method and single-ended fault location method. The influence of factors such as single-ended method grounding resistance on ranging accuracy is effectively eliminated by a double-ended method of ranging based on zero-sequence component^[4]. However, the double-ended method requires the installation of Phasor Measurement Unit (PMU), which can significantly increase the cost of grid operation. In addition, single-ended fault location suffers from fault initial angle and phase measurement problems and is strongly influenced by fault transition resistance and line parameters.

Some scholars have proposed to use the principle of parameter identification to overcome the problem of ranging error caused by transition resistance and line parameters^[10-12]. A double-ended ranging method based on PMU recording data was proposed in the literature^[11]. The method identifies the coupling parameters of the line by means of the voltage equation of the loop characterized with coupling parameters. The authors model the cable structure lines that correspond to the real situation in [12]. Then the zero-mode equivalent circuit based ranging equations are established to perform least squares parameter identification. Eventually the fault distance can be obtained.

In conclusion, this paper proposes a single-phase to ground fault location method based on the least-squares algorithm for arc-suppression-coil grounding system. On the basis of the faulty zero-sequence equivalent circuit, the fault location equation is constructed through the current-voltage relationship. The method obtains the fault distance by solving the constructed optimization model which considers the transition resistance, fault distance as unknown quantities. This method has the following advantages: a) It measures the electrical quantity of only one section of the line and does not require the installation of a synchronized measuring device; b) Instantaneous values are used in all calculations, so there is no need to measure fault initial angle and phase information; c) It overcomes the problem of the effect of transition resistance on ranging and has high ranging accuracy.

II. MODEL AND EQUATION OF FAULT LOCATION

A. Zero-sequence equivalent circuit

The most common topology in distribution network is the radial network. Figure 1 is the topology of radial distribution. There are n feeders in the network. When a single-phase to ground fault occurs on the n th feeder, its zero sequence equivalent circuit is shown in Fig. 2 [9-10].

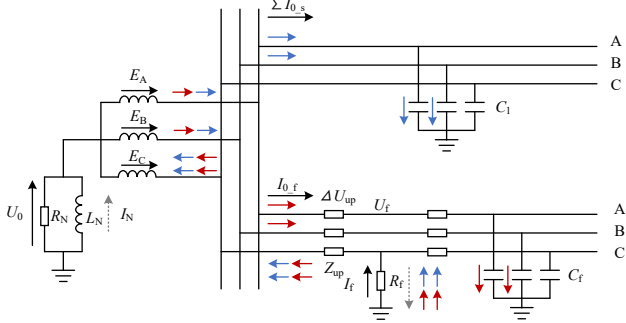


Fig. 1. Topology of radial network

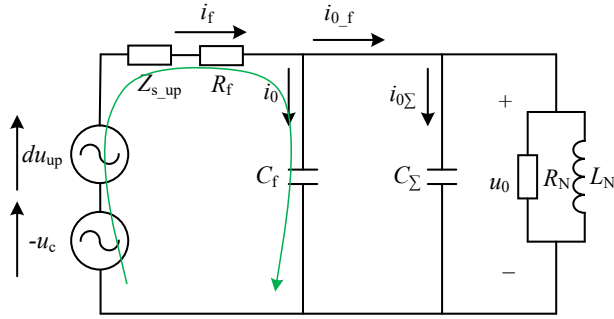


Fig. 2. Zero-sequence equivalent circuit of arc-suppression-coil earthed power distribution systems

This figure reflects the relationship between transition resistance, fault distance and current and voltage, and describes the fault characteristics of arc suppression coil grounding system when single-phase grounding fault occurs.

The fault distance is directly proportional to the upstream impedance of the fault. Hence, if the unit line resistance r_{s_up} and inductance l_{s_up} values are known, the fault distance d will be solved by upstream resistance through the fault. The fault upstream impedance Z_{s_up} is calculated as

$$Z_{s_up} = d(r_{s_up} + j\omega l_{s_up}) \quad (1)$$

B. Fault location equation

The KVL equation for the loop shown in Fig. 1 is established.

$$u_c - du_{up} = dr_{s_up}i_f + dl_{s_up}\frac{di_f}{dt} + R_f i_f + u_0 \quad (2)$$

The power supply voltage u_c and the zero-sequence voltage of bus u_0 can be measured. u_{up} is the voltage drop of the fault phase current of the faulty line on the upstream impedance per unit distance from the fault, which can be expressed by (3)

$$u_{up} = i_{cf}(r_{s_up} + j\omega l_{s_up}) \quad (3)$$

The relationship between fault current i_f , fault line zero sequence current i_{0_f} and fault capacitive current to ground i_0 exists as shown in (4)

$$i_f = i_{0_f} + i_0 \quad (4)$$

Which, i_{0_f} can be measured only. While i_0 is calculated through the Ohm's law in capacitance to earth C_f

$$i_0 = C_f \frac{du_0}{dt} \quad (5)$$

$$C_f = d \cdot c_f \quad (6)$$

Substituting (3)–(6) into (2)

$$\begin{aligned} & u_c - u_0 \\ &= d \left[r_{s_up} (i_{0_f} + dc_f \frac{du_0}{dt}) + l_{s_up} \frac{di_f}{dt} + u_{up} \right] \\ &+ R_f (i_{0_f} + dc_f \frac{du_0}{dt}) \end{aligned} \quad (7)$$

The differential in the above equation is converted to a difference quotient and then the equation is discretized.

$$\begin{cases} \frac{di_f}{dt} \approx \frac{i_f(k) - i_f(k-1)}{T} \\ \frac{du_0}{dt} \approx \frac{u_0(k) - u_0(k-1)}{T} \end{cases} \quad (8)$$

where k is the discrete point and T is the time interval. Applying (8) to (7), fault location equations is defined as

$$\begin{aligned} & u_c(k) - u_0(k) \\ &= d^2 \left[r_{s_up} \frac{c_f}{T} (u_0(k) - u_0(k-1)) \right. \\ &+ c_f \frac{l_{s_up}}{T^2} (u_0(k) - 2u_0(k-1) + u_0(k-2)) \left. \right] \\ &+ d \left[r_{s_up} i_{0_f}(k) + \frac{l_{s_up}}{T} (i_{0_f}(k) - i_{0_f}(k-1)) + u_{up}(k) \right] \\ &+ dR_f \frac{c_f}{T} (u_0(k) - u_0(k-1)) + R_f i_{0_f}(k) \end{aligned} \quad (9)$$

All variables in (9) are already known, except for the transition resistance and fault distance. Therefore, this equation is considered as the ranging equation to be solved.

The relationship between fault characteristics and fault parameters in zero-sequence equivalent circuits is derived, which is the fault location equation (9). In the next chapter, the algorithm to solve the fault distance by (9) is given.

III. ALGORITHMS OF FAULT LOCATION

A. The establishment of the objective function

The left side of the equal sign in (8) is the measurement of the voltage quantity, and the right side can be regarded as the calculation quantity. The error between computation and measurement at the k th sampling point is computed as

$$\begin{aligned} & \Delta u \\ &= u_c(k) - u_0(k) \\ &- d^2 \left[r_{s_up} \frac{c_f}{T} (u_0(k) - u_0(k-1)) \right. \\ &+ c_f \frac{l_{s_up}}{T^2} (u_0(k) - 2u_0(k-1) + u_0(k-2)) \left. \right] \\ &- d \left[r_{s_up} i_{0_f}(k) + \frac{l_{s_up}}{T} (i_{0_f}(k) - i_{0_f}(k-1)) + u_{up}(k) \right] \\ &- dR_f \frac{c_f}{T} (u_0(k) - u_0(k-1)) + R_f i_{0_f}(k) \end{aligned} \quad (10)$$

In theory, all sampling points should satisfy the condition of zero error, i.e.

$$F = \sum_k^n \Delta u^2 = 0 \quad (11)$$

However, (11) has two unknown parameters to be solved: the transition resistance R_f and the fault distance d . It cannot be solved by just one equation. So we need to approximate the true value by fitting algorithm. Therefore, the value of d when the function F obtains the minimum value is taken as the optimal solution of the function. The objective function of least squares is required to be

$$\min P = \frac{1}{2} F = \frac{1}{2} \sum_k^n \Delta u^2 \quad (12)$$

B. Least squares algorithms

There are two methods for solving least squares: direct derivation and iteration. For the nonlinear least squares model proposed in (12), the equation making the partial derivative of the objective function be zero is shown as

$$\begin{cases} \sum_k^n \Delta u \frac{\partial \Delta u}{\partial d} = 0 \\ \sum_k^n \Delta u \frac{\partial \Delta u}{\partial R_f} = 0 \end{cases} \quad (13)$$

Substituting (10) into the above equation, the system of equations containing the parameters to be solved is created as equation (14)

$$\begin{cases} \sum_k^n \Delta u \left[2d \left[r_{s_up} \frac{c_f}{T} (u_0(k) - u_0(k-1)) + c_f \frac{l_{s_up}}{T^2} (u_0(k) - 2u_0(k-1) + u_0(k-2)) \right] + \left[r_{s_up} i_{0_f}(k) + \frac{l_{s_up}}{T} (i_{0_f}(k) - i_{0_f}(k-1)) + u_{up}(k) \right] + R_f \frac{c_f}{T} (u_0(k) - u_0(k-1)) \right] \\ \sum_k^n \Delta u \left[-d \frac{c_f}{T} (u_0(k) - u_0(k-1)) + i_{0_f}(k) \right] = 0 \end{cases} \quad (14)$$

Equation 14 is a system of binary cubic equations for the transition resistance R_f and the fault distance d . The equation cannot be written directly as an analytical solution. Therefore, this paper utilizes numerical methods to approximate its solution.

In this paper, Newton's method is used to approximate the objective function in (12). The solution process is given an initial value, find increment a suitable descent method, and iterate. The most important step is to find the right direction to descend.

Performing the Taylor expansion of the objective function F in the neighborhood of the parameter variable $x = \{R_f, d\}$ is a straight forward descent method.

$$F(x + \Delta x) = F(x) + J^T \Delta x + \frac{1}{2} \Delta x^T H \Delta x \quad (15)$$

Where, J is the first order derivative of F with respect to x , i.e. the Jacobian matrix. H is the second order derivative of F with respect to x , i.e. the Hessian matrix. Our goal is to find the direction of decline. When the second-order term is retained, solving the incremental equation for the Newtonian descent direction obtained satisfies (16).

$$H(x) \Delta x = -J(x) \quad (16)$$

C. Process of algorithms

The fault location algorithm flow is shown in Figure 3 below:

1) Set system parameters. Determine the distribution network topology, each line length and impedance parameters.

2) Measure and process fault data. The zero sequence voltage of the fault bus and the zero sequence current of the fault line are measured and discretized.

3) Determine fault location equation. Calculate Jacobian matrix and Hessian matrix, and choice Newtonian descent direction.

$$J = \begin{bmatrix} \frac{\partial \Delta u}{\partial d} \\ \frac{\partial \Delta u}{\partial R_f} \end{bmatrix} = \begin{bmatrix} 2d \left[r_{s_up} \frac{c_f}{T} (u_0(k) - u_0(k-1)) + c_f \frac{l_{s_up}}{T^2} (u_0(k) - 2u_0(k-1) + u_0(k-2)) \right] + \left[r_{s_up} i_{0_f}(k) + \frac{l_{s_up}}{T} (i_{0_f}(k) - i_{0_f}(k-1)) + u_{up}(k) \right] + R_f \frac{c_f}{T} (u_0(k) - u_0(k-1)) - d \frac{c_f}{T} (u_0(k) - u_0(k-1)) + i_{0_f}(k) \end{bmatrix} \quad (17)$$

4) Set the initial value of the unknown parameter. Take $R_f = 0\Omega$ as the initial value of the transition resistance, and take half of the line length as the initial value of the fault distance.

5) Iterative calculation. When the increment is small enough or the number of iterations M exceeds the threshold M_{set} , the solution result is output.

$$M < M_{set} \quad (18)$$

The single-phase to ground fault point is set in the A phase of line 4.

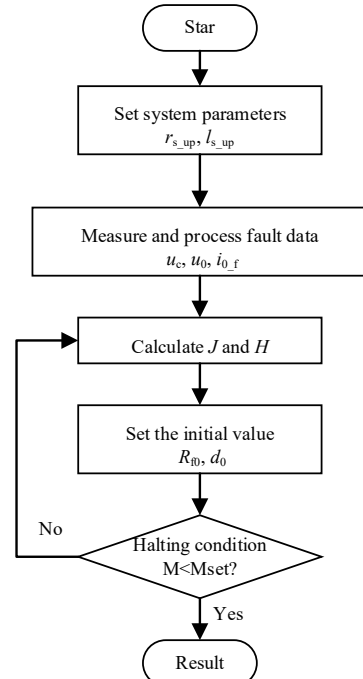


Fig. 3. Process of least squares algorithms

IV. SIMULATION

The simulation system is built in PSCAD, as shown in Figure 4. The system voltage level is 10kV. Arc suppression coil is grounded, and the compensation degree is 8%. The system has 4 outlet lines: 10km cable; Overhead line 9km; Cable 7km and overhead line 12km mixed; Cable 5km and overhead line 8km mixed.

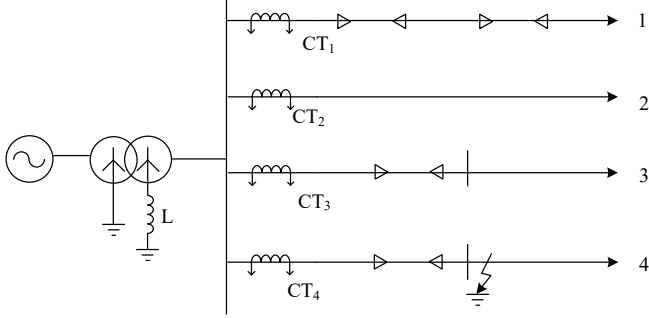


Fig. 4. Topology of simulation

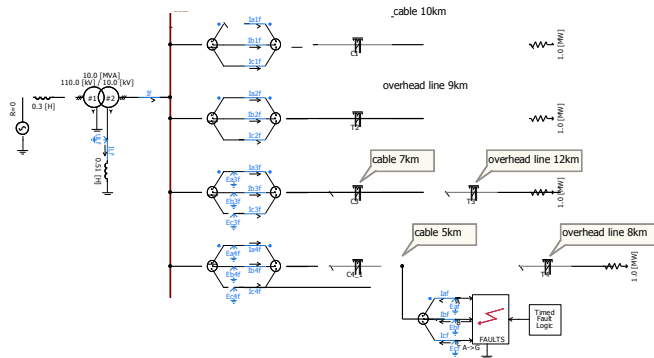


Fig. 5. Model of simulation

Parameters of overhead lines and cables are shown in Table 1 and 2 respectively.

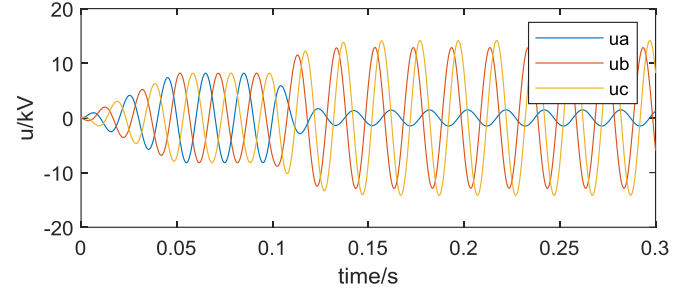
TABLE I. PARAMETERS OF LINES		
	positive sequence component	zero sequence component
Resistance Ω/km	0.17	0.23
Inductance mH/km	1.2096	5.4749
Capacitance $\mu\text{F}/\text{km}$	0.0097	0.006

TABLE II. PARAMETERS OF CABLES		
	positive sequence component	zero sequence component
Resistance Ω/km	0.27	2.7
Inductance mH/km	0.255	1.1019
Capacitance $\mu\text{F}/\text{km}$	0.3390	0.28

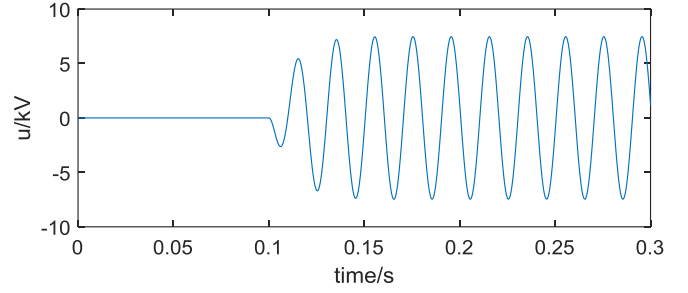
The single-phase to ground fault is set in the A phase of line 4 at tenth of a second after running. The data of 0.2s-0.22s is used for parameter identification. If the sampling interval is set, $n=20$ samples are taken.

Waveform of simulation is shown in Figure 6, where (a) shows the transformer outlet voltage; (b) shows the zero-sequence voltage of the bus; (c) shows the three-phase voltage of the fault line;

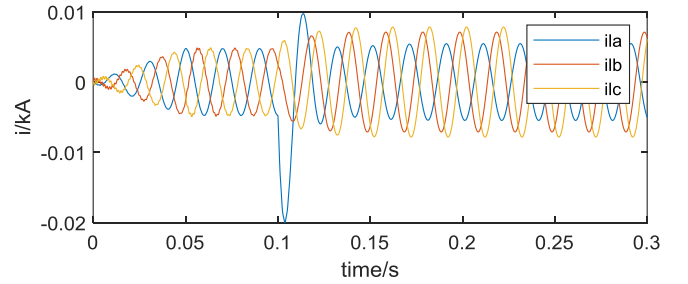
of the fault line, and (d) shows the zero-sequence current of the four feeders.



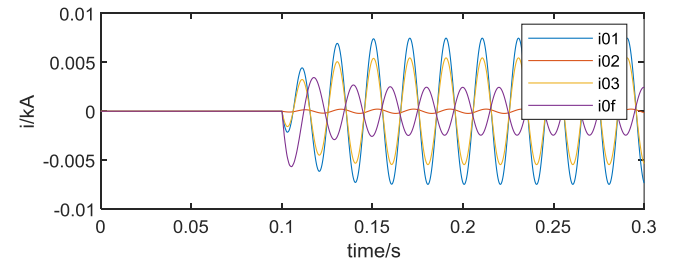
(a). the transformer outlet voltage



(b). the zero-sequence voltage of the bus



(c). the three-phase voltage of the fault line



d). the zero-sequence current of the four feeders

Fig. 6. Waveform of simulation

In order to verify the feasibility of the algorithm, it is necessary to measure and calculate multiple sets of data under different conditions. Set different fault locations and transition resistors showing as Table 1, and the identification results are shown in Table 2.

TABLE III. SETTING PARAMETERS	
d/km	Rf/ Ω
3	20
3	200
3	2000
7	20
7	200
7	2000

TABLE IV. RESULTS

d/km	R_f/Ω	errors%
3.354	18.76	11.8
2.826	187.2	5.8
3.357	1986.8	8.5
7.565	17.96	8.07
7.435	206.8	6.2
6.656	2008.4	4.9

It can be seen from Table 2 that the measurement error is within 12%, indicating that the proposed method has good feasibility.

V. CONCLUSION

The paper takes the single-phase to ground fault in arc-suppression-coil grounding systems as an example. The time-domain relationship of fault voltage and current is established as a distance measurement equation using the characteristic column of zero sequence current and voltage. On the premise that the measured values of the system and the calculated values of the mathematical model are known, the least squares model is established and the fault distance is obtained by Newton method. The method only needs to measure the electrical volume at one end of the line, and does not need to install a synchronous measuring device, which greatly reduces the installation cost. The instantaneous value is used to calculate the equation without measuring the initial fault Angle and phasor information. Taking the transition resistance as the unknown to calculate, the problem of the influence of the transition resistance on the distance measurement can be overcome, and higher ranging accuracy can be achieved.

VI. ACKNOWLEDGMENT

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